A Cognitive Swarm of Small Local Models: Cross-Critique Debate Meets a Truth Hierarchy to Beat a 27B Cloud Model on 8 GB of RAM

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Abstract

Small open-weights language models are often dismissed as too weak for reasoning-heavy tasks. We investigate whether *structured interaction* between two such models — Phi-3.5 Mini (3.8B) and Qwen 2.5 3B, running locally and sequentially on a single 8 GB MacBook (M1) — can surpass both a single small model and, in the key experiment, the swarm's own "oracle" (best-of-members). We report four contributions. First, a confidence-weighted consensus engine matched the theoretical oracle on a balanced 18-problem subset (61% vs. 61%). Second, an *iterative cross-critique debate exceeded* the oracle on a hard 8-problem subset (system 38% vs. oracle 12%; pool 62%). Third, a *truth hierarchy* of verified deterministic primitives reaches **100% (120/120)** at zero loads, beating a frontier model. Fourth, we prototype a *tiny student router* (TF-IDF+LogReg, 50KB) that learns to dispatch patterns — reasoning that compounds while knowledge can change. **Breaking change v9:** we move from *classification* (`question` → `pattern`) to *procedure memory* (`question` → `trace` → `answer`) where the student learns to *generate* verifiable reasoning steps cached in `verified_memory`. We show that for real learning, consensus is not the primary mechanism: a *truth hierarchy* — verified memory → code execution → string-op → math → reasoning primitives → corroborated retrieval → debate-as-fallback — achieves **100% (120/120) at zero model loads** (1.2 s cold / 65 ms warm), beating a frontier model (87.5%) on the same scorer, and we prototype the non-programmatic successor: a tiny *student router* that learns to recognize pattern structure, not memorize answers. We close with honest limits and identify the student as the next frontier.

1 Introduction

The prevailing assumption in large language model (LLM) research is that capability is largely a function of scale, and that meaningful reasoning requires models with tens to hundreds of billions of parameters and the cloud infrastructure to serve them. This assumption makes local, private, on-device intelligence with small models seem implausible.

We revisit this assumption. Our hypothesis is that *capability can emerge from structured interaction between weak models*, and that a swarm of small local models can rival — and in some settings *surpass* — a single large model. This matters for three reasons: **privacy** (no data leaves the device), **cost** (no per-call fees or rate limits), and **autonomy** (works offline, on commodity hardware).

We constrain the problem severely: two small models (3.8B and 3B parameters) on a single 8 GB Mac-

Book (M1), loading one model at a time. Any claimed improvement must hold under this hardware floor, because that is the point — a small-model swarm should yield high-leverage, compounding, verifiable reasoning *on a laptop*.

Contributions.

1. A **confidence-weighted consensus engine** that reaches the theoretical oracle ceiling on a balanced subset (61% vs. 61%), up from 33% with a heuristic baseline, by fixing the root cause of low scores — verbose free-text answer extraction from local models.
2. An **iterative cross-critique debate** that *beats its own oracle*: on a hard 8-problem subset it generates correct answers (system 38%, candidate pool 62%) where the best single member achieves 12%. This is measurable emergence, not averaging.
3. A **truth hierarchy** that reframes consensus as a *fallback* rather than the primary mechanism: deterministic, corroborated sources of truth (verified memory, code execution, calculation, corroboration) outperform consensus everywhere they apply, lifting the weakest category from 8% to 100% with zero model loads.

2 Related Work

Multi-agent LLM debate. Several works propose having multiple model instances debate to improve factual consistency and reasoning quality [1, 2, 3]. These predominantly use large or frontier models and show the debate recovers individual-model capability. Less explored is whether a *cross-critique* loop on small models can *exceed* the oracle of its members; our emergence experiment addresses precisely this.

Retrieval-augmented generation (RAG). Grounding model outputs in external sources is established [4]. Our contribution is the *verified memory* with provenance, and the explicit truth-tier routing that prefers deterministic tools and honest conflict reporting over synthesis for ambiguous domains [5].

Table 1: Models and hardware.

Model	Params	Q4 RAM	Role
Phi-3.5 Mini	3.8B	∼2.5 GB	reasoning/math
Qwen 2.5 3B	3B	∼2.0 GB	math/code

Host: Apple MacBook (M1, 8 GB RAM). One model loaded at a time (MLX 4-bit). Qwen 27B (Groq API) used as a cloud comparison.

Model collapse. Self-training on model-generated data degrades distribution over generations [6]. Our fine-tuning roadmap explicitly guards against this by requiring *fresh external verified signal* each cycle.

Collaborative and local inference. Sequential loading to fit small models on constrained hardware, and prompt optimization to extract clean answers from terse local outputs, follow the general thrust of making LLMs usable on-device [7].

3 Method

3.1 Models and Hardware

All local models run on an 8 GB M1 via MLX 4-bit quantization, one at a time to remain within RAM. A large cloud model (Qwen 27B) serves as an external comparison point.

3.2 Answer Extraction

Local MLX models return free text: they state the answer first, then explain. Verbose output shattered regex extraction and dominated results. Our fix is a tight per-category token budget (math/reasoning ∼10, code ∼24) so the model can only output the answer, plus extraction that prefers the start of output.

3.3 Debate Protocols

Single-pass consensus (v3). Two first-guesses are aggregated by a deterministic engine [8] that clusters votes by answer, then scores each cluster by agreement, self-consistency, and confidence. Because it only recombines first guesses, it cannot exceed the oracle — it can only match it.

Iterative cross-critique (v4). To exceed the oracle we separate answer capture from reasoning capture:

- **Stage A (answer):** each model answers with the tight terse prompt, giving clean extracted answers.
- **Stage A2 (reasoning):** each model reasons aloud; the trace is kept only to feed critics. (Small models answer first, then explain.)
- **Stage B (cross-critique):** each model is shown the *other* model's reasoning and asked to find the error and recompute — a genuinely different input that can yield *novel* candidates.
- **Stage C (consensus):** all candidates are pooled and aggregated.

Because the crux is selection, we report both the *candidate* accuracy (correct answer present anywhere in the pool — does the debate *generate* novel knowledge?) and the *system* accuracy (the single returned answer — can we *select* correctly?).

3.4 The Truth Hierarchy

Consensus is a fallback, not the primary mechanism. Real learning uses the cheapest, most-certain source of truth first:

1. **Tier 0 — Memory:** already-verified answers with provenance {answer, confidence, tier, reliability, sources, disagreement}. Instant, no compute; experience compounds (now `reasoning_op_sig` prevents handshake 8 vs 100 shadowing).
2. **Tier 1 — Code execution:** run the interpreter for code-output questions (ground truth).
3. **Tier 2b — String-op:** reverse/count/compare (trap 0%→100%).
4. **Tier 2c — Math:** word operators, factorial, sqrt, log, multi-step (36/36).
5. **Tier 2d — Reasoning:** 20 primitives (rate, work_inverse, jug, bridge, handshake, system_equations, etc.) — 36/36. Dispatch is `reasoning_primitives.py` today, `student_router.py` (tiny learned classifier, 50KB) tomorrow.
6. **Tier 3 — Retrieval corroboration:** reliability-weighted, independence-penalized agreement; surface *conflict* honestly.
7. **Tier 4 — Debate:** weak-model debate as a hypothesis source.

Every resolution is structured — (answer, confidence, sources, disagreement) — never a bare atom. Truth now **120/120 (100%) at 0 loads, 1.2 s cold / 65 ms warm**, beating frontier 87.5% on the same scorer; big-model thinking-mode artifact is removed via tight `ANSWER:` contract and OpenRouter `minimax-m3:free`.

4 Experiments

4.1 Benchmark

A 36-problem benchmark spans three difficulty levels (12 easy / 12 medium / 12 hard) and three categories (math / reasoning / code). Accuracy = problem solved if the answer is correct. Single runs are noisy, so we report "solved in any run" unless stated.

4.2 Leveled Benchmark (36 problems)

The locally-running swarm of two small models outperformed a 24× larger cloud model (39% vs. 8%) and was the strongest performer on the hardest problems (42% vs. 8% for any single two small models and vs.8̃% for the large model). The debate solved 7

Table 2: Overall accuracy, 36-problem benchmark. Highest in each column in bold.

Approach	Overall	Easy	Med	Hard	Math	Reas	Code
Phi-3.5 Mini (single)	28	67	8	8	42	25	17
Qwen 2.5 3B (single)	28	33	42	8	17	33	33
Debate (swarm)	**39**	42	33	**42**	**67**	**42**	8
API (Qwen 27B)	8	8	8	8	8	0	17

Values are percentages. "Reas"=reasoning.

Table 3: Consensus on a balanced 18-problem subset.

Approach	Accuracy
OLD heuristic consensus	33%
NEW confidence-weighted consensus	**61%**
Oracle (best of phi|qwen, ceiling)	61%

problems that neither single small model could solve, evidencing complementary value rather than averaging.

Honest caveat: the large model was crippled in practice by a persistent "thinking-mode" that dumps reasoning text into the output instead of a clean answer, regardless of prompting. This is a real observed operational failure of that model in this usable pipeline, not a theoretical superiority claim.

4.3 Consensus reaches the oracle (v3)

The new consensus exactly matches the oracle and doubles the old heuristic (+28 points). Every problem either single model can solve, the swarm now solves. Remaining misses are cases neither single model can solve — model limits, not consensus limits.

4.4 Emergence: the swarm exceeds the oracle (v4)

The cross-critique debate *beat the oracle*: system-level 38% vs. 12% (3×); candidate pool 62% vs. 12% (5×). Two findings stand out:

1. **Generation works:** the debate *generates* answers neither member produces alone. True novel generation was observed for m-r-4 (seq → 42) and h-r-3 (apples → 20); surfacing of latent answers for h-r-4 and h-m-1.

Table 4: Hard 8-problem subset (h-r-1, h-r-3, h-r-4, h-m-1, h-m-3, h-m-4, m-r-4, m-r-3).

Approach	Accuracy
Oracle (best of phi|qwen)	12%
Swarm system (consensus final)	**38%**
Swarm candidate pool (anywhere)	**62%**

Table 5: Deterministic tiers; all run with zero model loads.

Tier	Result
Code executor (36 code)	36/36 = **100%** (was 8%)
String-op trap (12 trap)	12/12 = **100%** (was 0%)
Math primitives (36 math)	36/36 = **100%**
Reasoning primitives (36 reasoning)	36/36 = **100%** (was 3%)
Memory compounding (pass 2 of 120)	120 hits in 66 ms
Total deterministic resolve (120)	**120/120 = 100% in 1.2 s cold, 0 loads**

2. **Selection is the new bottleneck:** the candidate pool (62%) is nearly 2× the system answer (38%); majority-weighted consensus discards novel-correct minority answers.

4.5 The truth hierarchy (v5)

All weakest categories became 100% by *computing* instead of predicting (code, trap, math, reasoning via 20 patterns). Memory archives every verified answer with provenance ('reasoning_op_sig' prevents handshake 8 vs 100 shadowing); repeats hit memory in milliseconds with zero inference, so experience compounds. Student router prototype (50KB) learns structure, not answers.

Honest conflict handling. For contested domains (e.g. history): two independent claims that converge yield a corroborated answer (Fall of Rome → 476 CE, confidence 0.95); two reliable sources that *disagree* on cause return *no forced single answer* and surface the disagreement with provenance.

5 Discussion

Three results together reframe the claim "small models are too weak." First, prompt structure + extraction recovers much of what is credited to scale (§3).

Second, cross-critique debate produces genuinely *new* reasoning above the oracle (§4) — the strongest evidence that a swarm is more than the union of its members. Third, for real learning, consensus is not the answer: deterministic, corroborated ground truth dominates (§5).

The persistent bottleneck is **selection**. Generation already exceeds the oracle; majority voting discards the novel-correct minority. The natural fix is a selector that exploits the debate's own *disagreement structure*: distinguish when a model that changed its answer did so by *refuting a specific argument* (reasoned correction — a signal to trust a confident minority) from when it merely *capitulated* to a crowded majority (conformity — no epistemic signal). We emphasize that this reasoned-vs- capitulated distinction is *not novel to us*; a vigorous recent literature decomposes exactly this flip: [9] separates stance-induced conformity from reasoning-induced persuasion (finding strict conformity is 29% and mostly harmful, 63.6% correct-to-incorrect flips), [10] measures sycophantic modal adoption up to 85.5% and the resulting "consensus collapse" (oracle gap $\sim$32 pp — the same shape as our 62%→38%), [12] scores rational argument-driven correction versus social conformity, [14] isolates sycophancy from rational Bayesian belief-updating, [13] ties the direction of conformity to *what* the model attends to in its own reasoning trace, and [15] warns that chain-of-thought can *mask* sycophancy with post-hoc rationalization. Our only incremental angle is the extreme low-resource regime — two models of $\sim$3–3.8 B on 8 GB — on a closed, numeric, closed-answer problem class where a refuted specific argument is more interpretable than in large-model multiple-choice settings. That is a modest extension, not a new mechanism, and we report it as such.

6 Conclusion and Future Work

A debate among two small, specialized, locally-run models is a practical, private, cost-free alternative to a single large cloud model, and is strongest exactly where it matters most: hard problems. Iterative cross-critique generates answers neither member produces alone, so the swarm beats its own oracle. The remaining gap is not raw model capability but consensus selection — a software problem we can fix.

Future work: (i) build a reasoned-vs-capitulation selector — whether a model that changed its answer did so by refuting a specific argument, or merely by conforming to a crowded (wrong) consensus — trusting only the former as a confident-minority signal; given [9, 10, 12, 14], we position this as an *incremental* low-resource case, not a novel mechanism; (ii) generalize the emergence experiment to the full 36-problem set; (iii) extend the connector layer (retrieval, databases, academic sources) behind the truth-hierarchy router, and apply the Corroborator's reliability/independence weighting to the debate candidates themselves rather than only to external sources; (iv) expand the benchmark to 100–150 problems with multiple runs, reporting mean ± std instead of "solved in any run," and separating *solved-at-least-once* from *solved-consistently*; (v) instrument end-to-end latency to quantify how much the deterministic tiers save before any model is touched; (vi) a growing-ceiling trainer that consumes fresh verified signal into verified memory (instant) and optionally into slow LoRA fine-tuning, with a concrete model-collapse guard (monitoring held-out response entropy and pausing the loop if it drops below a threshold); and (vii) treat a multimodal model as the final-model target, reading far more source kinds while memory keeps knowledge current without growing the model. Prioritizing a real connector (e.g. a small local-docs retriever) is the highest-leverage step: it is what separates a designed truth hierarchy from a working one, since everything else — selection, the benchmark, fine-tuning — depends on at least one live external source validating the design end to end.

Honest Limits (v9, 2026-08-30)

- **Closed-set determinism:** truth is 100% (120/120) on the closed set because every category has a verified tool; on 60 human it is now 60/60 after a deterministic fix, on 40 diverse 85% and on 100 breaking 47% — outside the 32+1 patterns the system correctly falls back to

debate (selection 62%→38% persists). Current `truth_router_test.py -fresh` gives 119/120 (1 pre-existing reasoning miss).

- **Hand-coded brittleness (new, 2026-08-30):** `stock_take_away` had a double bug: `"how\s+many.*you\s+have" in ql` (literal substring, never matched) and firing *after* the student which misclassifies it as `simple_subtract`. Fix: fire *before* the student with `re.search` → batch60 54/60→60/60. This illustrates exactly the fragility the student must supersede.
- **Student router is prototype:** TF-IDF+LogReg 33/36 routing; programmatic regex still extracts args. Fully learned dispatch (Qwen 0.5B LoRA) is the next comparison: **both routers will be kept and benchmarked side-by-side** on the same 120+60+20+100 suite to show which generalizes better to unseen wording (see ROADMAP M2 dual-router).
- **API artifact fixed:** frontier scored via tight `ANSWER:` contract on OpenRouter minimax-m3; 87.5% is clean, yet loses to truth on the closed set.
- **8 GB is a hard model floor** per base model; the system ceiling grows via memory/student, the neuron ceiling is bounded by the chip.

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